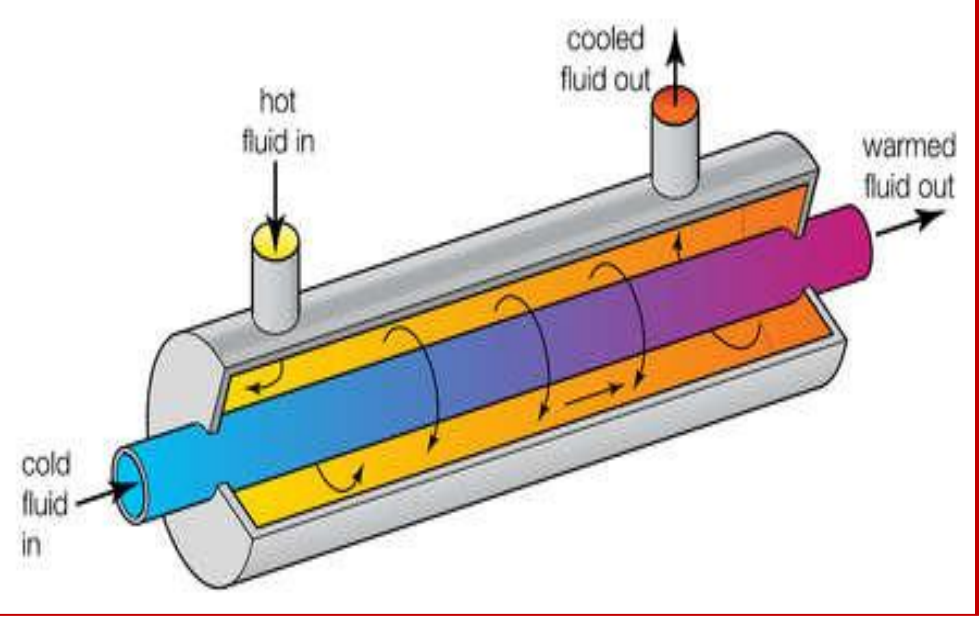


# Problem Set Reflection Homework Assignment

## Summary

As part of an assignment for my Heat Transfer class, I was responsible for reflecting upon one of my past problem sets. For this reflection, I decided to revisit my third problem set of the semester, where we were beginning to utilize the concepts of nonlinear heat flow and differential equations. I had received 94.5 percent on this assignment, so I will also reflect upon the mistake I had made and what I've learned from it as well.

## Reflection

The problem set was made up of three questions. The first focused on analyzing a heat transfer problem as a control volume, teaching us to utilize energy balance as a differential equation to solve for temperature. The second question was about using various boundary conditions within a multilayer system to solve for the temperature at different points. We could then use this information to understand that the conduction through different layers took a nonlinear form when heat was being generated. And the final question regarded the application of a ceramic coating on a propeller blade. Here, we needed to solve for the initial max temperature that the blade would undergo without the coating and then find the new maximum temperature once the coating was applied. This made us use a new concept we had learned, heat resistances, which I found was a remarkable similarity to the circuit knowledge I'm also learning in my mechatronics class.

The mistake I had made was in the first problem. I had correctly identified the need for an energy balance, using it to create a second order differential equation that could be solved to find an accurate value for temperature within the control volume that was provided. I then solved this differential equation, then identified the boundary conditions. However, I made the mistake of assuming the system was in steady state. This led me to have the incorrect differential equation when solving, which marked me off some points.

Name: Hugo Larios hd145 Date of Submission: 2/24/2026

Problem set number: #3 Due Date: 2/24/2026

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Problem 1)

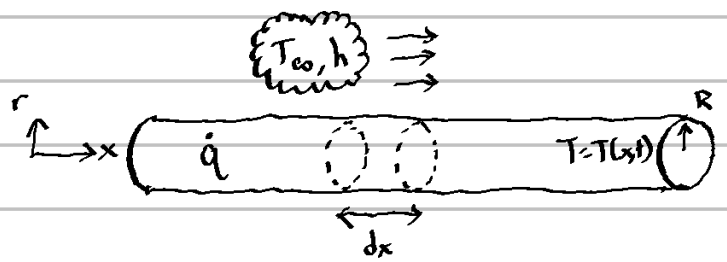
Known: Consider a long rod w/ radius $R$ and a uniform volumetric heat generation rate of $\dot{q}$. Rod is placed in an environment w/ ambient temperature $T_\infty$ and heat transfer coefficient $h$. The x-axis is set through the axis of the rod. The temperature of the rod is a function of $x$ and time $t$, e.g., $T = T(x,t)$.

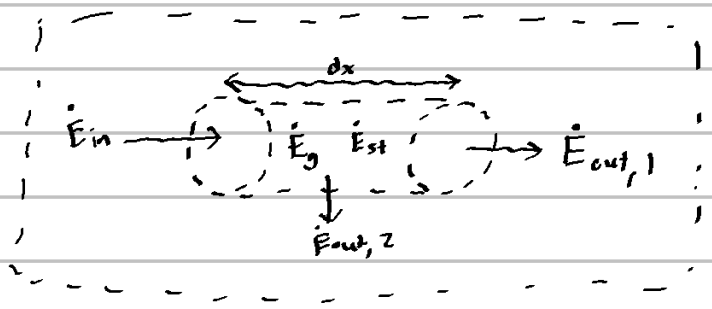
Find: a) Considering a differential cylinder w/ a height of $dx$ as the CV, write down the energy balance equation for the control volume. Then express each energy term in the energy balance equation as a function of $T$.

b) Derive the governing differential equation that describes the temperature profile of the rod and simplify.

c) Provided that the rod has a length $L$, $T(0,t) = T_1$, $T(L,t) = T_\infty$, and $T(x,0) = T_i$, write the proper boundary conditions and initial conditions for the governing differential equation.

Schematic:



Properties: Conduction w/ $k$, Convection w/ $h$, Heat generation $\dot{q}$

Assumptions: 1) Thermal Radiation is negligible; 2) The radius of rod is very small; 3) Temperature of the rod along the radial direction is uniform. 4) Assume constant $k$, $\rho$, and $c_p$ 5) Assume steady-state

Analysis: a) We start by writing down a starting energy balance equation and substituting with our knowns.

$$\dot{E}_{in} - \dot{E}_{out} + \dot{E}_g = \cancel{\dot{E}_{st}} = 0 \rightarrow \dot{q}_{cond,in} - \dot{q}_{cond,out} - \dot{q}_{conv} + \dot{q} = 0$$

$$q_x - \left(q_x + \frac{dq_x}{dx}dx\right) - hA_e\,dx\,(T - T_\infty) + \dot{q}A_i\,dx = 0$$

$$\boxed{-kA_i\frac{dT}{dx} - \left(-kA_i\frac{dT}{dx} - kA_i\frac{d^2T}{dx^2}dx\right) - hA_e\,dx(T-T_\infty) + \dot{q}A_i\,dx = 0}$$

b) $$\cancel{q_x} - \left(\cancel{q_x} + \frac{dq_x}{dx}dx\right) - hA_e\,dx(T-T_\infty) + \dot{q}A_i\,dx = 0$$

$$-\frac{d}{dx}q_x\,\cancel{dx} - hA_e\,\cancel{dx}\,(T-T_\infty) + \dot{q}A_i\,\cancel{dx} = 0$$

$$kA_i\frac{d^2T}{dx^2} - hA_e(T-T_\infty) + \dot{q}A_i = 0$$

$$k\pi R^2\frac{d^2T}{dx^2} - 2\pi Rh(T-T_\infty) + \dot{q}\pi R^2 = 0$$

$$kR\frac{d^2T}{dx^2} - 2h(T-T_\infty) + \dot{q}R = 0$$

$$\boxed{\frac{kR}{2h}\frac{d^2T}{dx^2} - T = -T_\infty - \frac{\dot{q}R}{2h}}$$

c) $$\boxed{T(x=0,t) = T_1 \qquad T(x=L,t) = T_\infty \qquad T(0<x<L,\ t=0) = T_i}$$

Problem 2)

Known: We have a plane composite wall separated into Wall A & Wall B. Wall A has a thickness of $L_A = 30$ cm, which experiences uniform volumetric heat generation $\dot{q}$. The thermal conductivity of wall A is $k_A$. The left surface of wall A is thermally insulated. Wall B has a thickness of $L_B = 10$ cm and thermal conductivity $k_B = 30$ W/(m·K). There is no heat generation in wall B. The left surface of wall B is in contact w/ wall A, w/ no contact thermal resistance at their boundary. The right surface of wall B is exposed to a coolant at $T_\infty = 20°C$ and heat transfer coefficient $h = 500$ W/(m²·K). A sensor between wall A and the insulation shows $T_w = 115°C$ and another sensor between walls A and B reads $T_i = 100°C$

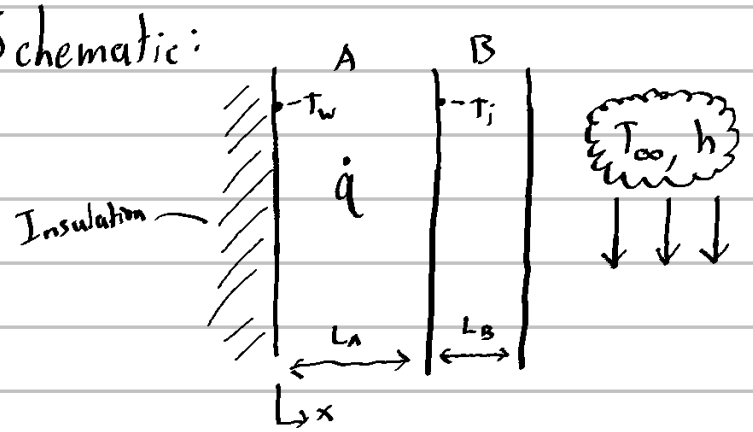
Find: a) The right surface temperature $T_s$ of wall B.

b) The uniform volumetric heat generation $\dot{q}$ at steady state.

c) The thermal conductivity $k_A$ across wall A.

d) Plot the temperature profile across wall A and wall B and show its important features.

Schematic:



Properties: Conduction w/ $k_A, k_B$; Convection w/ $h$; Heat generation w/ $\dot{q}$; thermal insulator boundary

Assumptions: 1) 1D Heat Conduction; 2) Uniform volumetric heat generation; 3) Negligible heat radiation. 4) Assume steady state.

Analysis: a) Because area is not provided, we can use heat flux to determine $T_s$.

$$\dot{E}_{in} + \cancel{\dot{E}_g} - \dot{E}_{out} = \cancel{\dot{E}_{st}} \rightarrow \dot{E}_{in} - \dot{E}_{out} = 0$$

$$q''_{cond_B} = q''_{conv_B} \rightarrow -k_B \frac{T_s - T_i}{L_B} = h(T_s - T_\infty)$$

$$T_s - T_i = -\frac{hL_B}{k_B} T_s + \frac{hL_B}{k_B} T_\infty$$

$$T_s\left(\frac{hL_B}{k_B} + 1\right) = \frac{hL_B}{k_B} T_\infty + T_i$$

$$T_s = \frac{\frac{hL_B}{k_B} T_\infty + T_i}{\frac{hL_B}{k_B} + 1} = 323.15\ K \boxed{= 49.85°\ C}$$

b) We can set up an energy balance equation to solve for $\dot{q}$. Because the left surface is insulated and the right surface has heat flow toward wall B, we can have $\dot{E}_{in} = 0$ and $\dot{E}_{out} = q''_{cond,B}$.

$$\cancel{\dot{E}_{in}} + \dot{E}_g - \dot{E}_{out} = \cancel{\dot{E}_{st}} \rightarrow \dot{E}_g = \dot{E}_{out}$$

$$\dot{q} L_A = q''_{cond,B} = -k_B \frac{T_s - T_i}{L_B}$$

$$\dot{q} = -k_B \frac{T_s - T_i}{L_B L_A} = -(30)\frac{49.85 - 100}{(0.3)(0.1)} \boxed{\approx 50,150\ W/m^3}$$

c) We can use Fourier's Law by utilizing the heat generation/heat flux we found in b).

$$k_A \frac{\partial^2 T}{\partial x^2} + \dot{q} = 0 \rightarrow \frac{\partial^2 T}{\partial x^2} = -\frac{\dot{q}}{k_A}$$

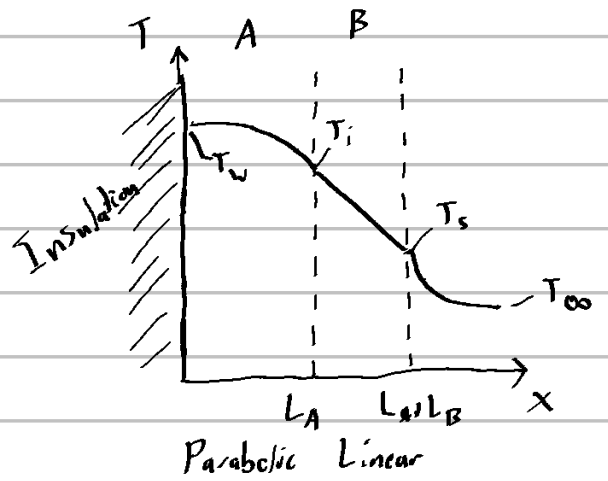
$$T(x) - \frac{\dot{q}}{2k_A} x^2 + ax + b \qquad \frac{\partial T}{\partial x} = 0 \Big|_{x=0}$$

$$T(0) = 155$$

$$T(L_A) = T_i = \frac{-\dot{q} L_A^2}{2k_A} + 155$$

$$k_A = \frac{-\dot{q} L_A^2}{2(T_i - 155)} \approx \frac{-50{,}150\,(0.3)^2}{2(100-115)} \boxed{\approx 150.45\ W/_{m \cdot K}}$$

d) Plot the temperature profile



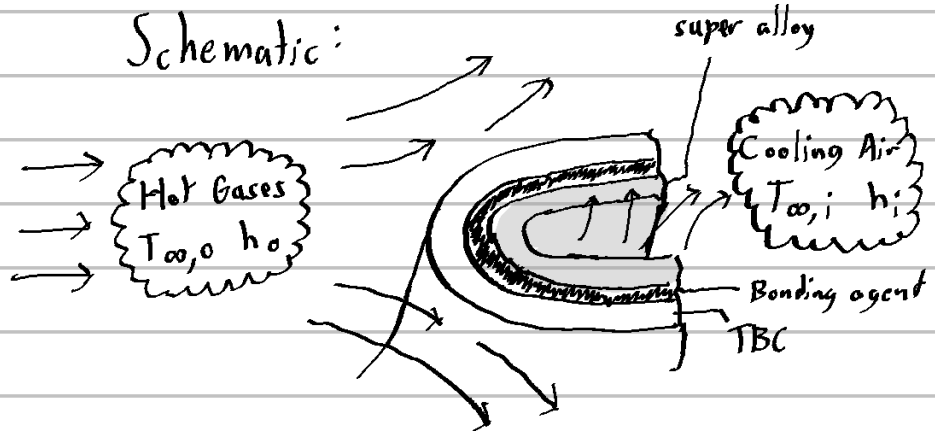
Problem 3)

Known: A turbine blade is made of a high-temperature superalloy w/ thermal conductivity of $k_b = 25$ W/(m·K). It is coated by a ceramic material w/ thermal conductivity of $k_b = 1$ W/(m·K). In order to attach the coating, a metallic bonding agent is needed, which creates an interfacial thermal resistance (per unit area) of $R'' = 10^{-4}$ m²·K/W. Consider the operating condition for which external hot gases at $T_{\infty,o} = 1616$ K and cooling air inside the blade at $T_{\infty,i} = 400$ K. Heat transfer coefficients are $h_o = 1000$ W/(m²·K) and $h_i = 500$ W/(m²·K). The thickness of the superalloy blade is $L_b = 5$ mm. The max allowable temperature is $T_{max} = 1,200$ K.

Find: a) Draw the thermal resistance network and label all thermal resistances when the TBC is not applied. Calculate the exterior surface temperature of the superalloy blade. Can it be maintained below $T_{max}$?

b) Draw the thermal resistance network and label all thermal resistances when $L_{TBC} = 0.5$ mm thick TBC is applied. Calculate the temperature of the superalloy blade. Can the blade be maintained below $T_{max}$ this time?

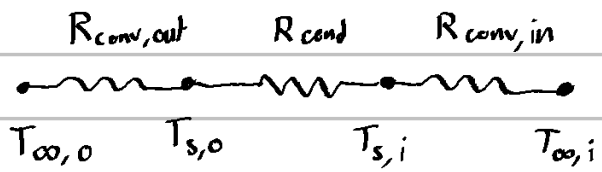
Schematic:




Properties: Conduction, $k_b$, $k_{TBC}$, $R''$
Convective Fluid Flow, $h_o$, $h_i$

Assumptions: 1) Thickness of bonding agent is negligible; 2) Steady-state condition; 3) Thermal radiation is negligible; 4) Turbine blade can be approximated as a plane wall; 5) 1D heat conduction

Analysis: a)




$R_{conv,out} = \frac{1}{h_o}$

$R_{cond} = \frac{L_B}{k_b}$

$R_{conv,in} = \frac{1}{h_i}$

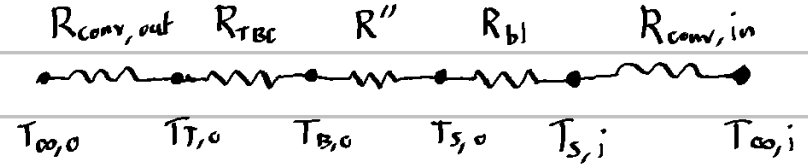
$$T_{\infty,o} - T_{\infty,i} = q'' R_{eff} \qquad R_{eff} = \frac{L}{k_B} + \frac{1}{h_o} + \frac{1}{h_i} \approx 3.2\times10^{-3}$$

$$q'' = \frac{T_{\infty,o} - T_{\infty,i}}{R_{eff}} = \frac{1616-400}{0.0032} \approx 380 \ kW/m^2$$

$$q'' = \frac{T_{\infty,o} - T_{s,o}}{R_{conv,out}} \rightarrow -R_{conv}\, q'' + T_{\infty,o} = T_{s,o} \boxed{\approx 1{,}236 \ K}$$

1236 K > 1200 K, so no, the blade cannot be maintained below $T_{max}$

b)




$R_{conv,out} = \frac{1}{h_o} = 0.001$ $\quad R_{TBC} = \frac{L_{TBC}}{k_{TBC}} \approx 5\times10^{-4}$

$R'' = 10^{-4}$ $\quad R_{bl} = \frac{L_B}{k_B} = 2\times10^{-4}$ $\quad R_{conv,in} \approx 0.002$

$T_{\infty,out} - T_{\infty,in} = R_{eff}\, q''$ $\quad R_{eff} = R_{conv,out} + R_{TBC} + R'' + R_{bl} + R_{conv,in}$

$\approx 3.8\times10^{-3}$

$$q'' = \frac{T_{\infty,out} - T_{\infty,in}}{R_{eff}} = \frac{1616-400}{3.8\times10^{-3}} \approx 320 \text{ kW/m}^2$$

$$q'' = \frac{T_{\infty,out} - T_{S,o}}{R_{conv,out} + R_{TBC} + R''}$$

$$T_{S,o} = T_{\infty,out} - (R_{conv,out} + R_{TBC} + R'')\, q''$$

$$\boxed{\approx 1104 \text{ K}}$$

$1104 \text{ K} < T_{max} = 1200 \text{ K}$, so we can operate the blade now.